

结对作业使用说明 10.01更新

elevator-py最后更新时间：2025.10.01 18:12

1. 查阅模拟器项目（`pip install elevator-py` 将于已在 10月1日 上线）

开源地址：<https://z.gitee.cn/zgca/repos/zgca/Elevator>

文档地址：<https://zgca-forge.github.io/Elevator>

2. 安装测试框架



Copy

直接安装

```
pip install elevator-py
```

网络不通可使用使用清华源下载，指令如下：

```
pip install elevator-py -i
```

```
https://mirrors.tuna.tsinghua.edu.cn/pypi/web/simple
```

3. 运行电梯模拟服务



Copy

默认会在 `http://127.0.0.1:8000` 开启服务，如果无法运行，请检查8000端口是否被占用

```
python -m elevator_saga.server.simulator
```

4. 运行电梯客户端



Copy

```
python xxxx.py # 运行你的python文件即可
```

客户端示例

Copy

```
#!/usr/bin/env python3
```

```
from typing import List
```

```
from elevator_saga.client.base_controller import ElevatorController
```

```
from elevator_saga.client.proxy_models import ProxyElevator,  
ProxyFloor, ProxyPassenger
```

```
from elevator_saga.core.models import SimulationEvent
```

```
class TestElevatorBusController(ElevatorController):
```

```
    def __init__(self):
```

```

super().__init__("http://127.0.0.1:8000", True)

def on_init(self, elevators: List[ProxyElevator], floors:
List[ProxyFloor]) -> None:
    pass

def on_event_execute_start(
    self, tick: int, events: List[SimulationEvent], elevators:
List[ProxyElevator], floors: List[ProxyFloor]
) -> None:
    pass

def on_event_execute_end(
    self, tick: int, events: List[SimulationEvent], elevators:
List[ProxyElevator], floors: List[ProxyFloor]
) -> None:
    pass

def on_passenger_call(self, passenger: ProxyPassenger, floor:
ProxyFloor, direction: str) -> None:
    pass

def on_elevator_idle(self, elevator: ProxyElevator) -> None:
    pass

def on_elevator_stopped(self, elevator: ProxyElevator, floor:
ProxyFloor) -> None:
    pass

def on_passenger_board(self, elevator: ProxyElevator, passenger:
ProxyPassenger) -> None:
    pass

def on_passenger_alight(self, elevator: ProxyElevator, passenger:
ProxyPassenger, floor: ProxyFloor) -> None:
    pass

def on_elevator_passing_floor(self, elevator: ProxyElevator, floor:
ProxyFloor, direction: str) -> None:
    pass

def on_elevator_approaching(self, elevator: ProxyElevator, floor:
ProxyFloor, direction: str) -> None:
    pass

```

```
if __name__ == "__main__":
    algorithm = TestElevatorBusController()
    algorithm.start()
```

5. 乘客生成文件 `elevator_saga/traffic`

```
{
  "building": {
    "floors": 6,
    "elevators": 2,
    "elevator_capacity": 8,
    "scenario": "inter_floor",
    "scale": "medium",
    "description": "楼层间流量 - 适合小建筑 (medium规模)",
    "expected_passengers": 60,
    "duration": 200
  },
  "traffic": [
    {
      "id": 1,
      "origin": 5,
      "destination": 1,
      "tick": 0
    },
    {
      "id": 2,
      "origin": 5,
      "destination": 1,
      "tick": 5
    },
    {
      "id": 3,
      "origin": 1,
      "destination": 5,
      "tick": 9
    },
    {
      "id": 4,
      "origin": 1,
      "destination": 3,
      "tick": 11
    },
    {
      "id": 5,
```

```
"origin": 5,  
"destination": 1,  
"tick": 13  
},  
{  
  "id": 6,  
  "origin": 2,  
  "destination": 3,  
  "tick": 17  
},  
{  
  "id": 7,  
  "origin": 3,  
  "destination": 2,  
  "tick": 18  
},  
{  
  "id": 8,  
  "origin": 4,  
  "destination": 3,  
  "tick": 19  
},  
{  
  "id": 9,  
  "origin": 4,  
  "destination": 1,  
  "tick": 20  
},  
{  
  "id": 10,  
  "origin": 4,  
  "destination": 1,  
  "tick": 21  
},  
{  
  "id": 11,  
  "origin": 3,  
  "destination": 1,  
  "tick": 22  
},  
{  
  "id": 12,  
  "origin": 4,  
  "destination": 3,  
  "tick": 24
```

```
},
{
  "id": 13,
  "origin": 4,
  "destination": 1,
  "tick": 25
},
{
  "id": 14,
  "origin": 4,
  "destination": 3,
  "tick": 26
},
{
  "id": 15,
  "origin": 4,
  "destination": 2,
  "tick": 28
},
{
  "id": 16,
  "origin": 3,
  "destination": 2,
  "tick": 29
},
{
  "id": 17,
  "origin": 5,
  "destination": 3,
  "tick": 34
},
{
  "id": 18,
  "origin": 3,
  "destination": 1,
  "tick": 36
},
{
  "id": 19,
  "origin": 3,
  "destination": 2,
  "tick": 38
},
{
  "id": 20,
```

```
"origin": 2,  
"destination": 5,  
"tick": 43  
},  
{  
  "id": 21,  
  "origin": 3,  
  "destination": 2,  
  "tick": 46  
},  
{  
  "id": 22,  
  "origin": 4,  
  "destination": 1,  
  "tick": 51  
},  
{  
  "id": 23,  
  "origin": 5,  
  "destination": 4,  
  "tick": 52  
},  
{  
  "id": 24,  
  "origin": 5,  
  "destination": 1,  
  "tick": 54  
},  
{  
  "id": 25,  
  "origin": 1,  
  "destination": 4,  
  "tick": 58  
},  
{  
  "id": 26,  
  "origin": 4,  
  "destination": 3,  
  "tick": 62  
},  
{  
  "id": 27,  
  "origin": 2,  
  "destination": 3,  
  "tick": 63
```

```
},
{
  "id": 28,
  "origin": 2,
  "destination": 4,
  "tick": 66
},
{
  "id": 29,
  "origin": 3,
  "destination": 5,
  "tick": 68
},
{
  "id": 30,
  "origin": 4,
  "destination": 1,
  "tick": 69
},
{
  "id": 31,
  "origin": 1,
  "destination": 2,
  "tick": 71
},
{
  "id": 32,
  "origin": 1,
  "destination": 3,
  "tick": 79
},
{
  "id": 33,
  "origin": 5,
  "destination": 1,
  "tick": 81
},
{
  "id": 34,
  "origin": 4,
  "destination": 1,
  "tick": 83
},
{
  "id": 35,
```

```
"origin": 5,  
"destination": 2,  
"tick": 90  
},  
{  
  "id": 36,  
  "origin": 2,  
  "destination": 1,  
  "tick": 93  
},  
{  
  "id": 37,  
  "origin": 2,  
  "destination": 4,  
  "tick": 94  
},  
{  
  "id": 38,  
  "origin": 2,  
  "destination": 3,  
  "tick": 102  
},  
{  
  "id": 39,  
  "origin": 5,  
  "destination": 1,  
  "tick": 105  
},  
{  
  "id": 40,  
  "origin": 4,  
  "destination": 5,  
  "tick": 106  
},  
{  
  "id": 41,  
  "origin": 3,  
  "destination": 4,  
  "tick": 109  
},  
{  
  "id": 42,  
  "origin": 4,  
  "destination": 3,  
  "tick": 122
```



```
},
{
  "id": 43,
  "origin": 5,
  "destination": 2,
  "tick": 127
},
{
  "id": 44,
  "origin": 5,
  "destination": 2,
  "tick": 130
},
{
  "id": 45,
  "origin": 1,
  "destination": 2,
  "tick": 135
},
{
  "id": 46,
  "origin": 5,
  "destination": 3,
  "tick": 139
},
{
  "id": 47,
  "origin": 1,
  "destination": 5,
  "tick": 144
},
{
  "id": 48,
  "origin": 5,
  "destination": 2,
  "tick": 146
},
{
  "id": 49,
  "origin": 3,
  "destination": 4,
  "tick": 151
},
{
  "id": 50,
```

```
"origin": 2,  
"destination": 4,  
"tick": 155  
},  
{  
  "id": 51,  
  "origin": 3,  
  "destination": 5,  
  "tick": 157  
},  
{  
  "id": 52,  
  "origin": 5,  
  "destination": 2,  
  "tick": 163  
},  
{  
  "id": 53,  
  "origin": 3,  
  "destination": 1,  
  "tick": 172  
},  
{  
  "id": 54,  
  "origin": 2,  
  "destination": 4,  
  "tick": 174  
},  
{  
  "id": 55,  
  "origin": 4,  
  "destination": 1,  
  "tick": 179  
},  
{  
  "id": 56,  
  "origin": 3,  
  "destination": 4,  
  "tick": 186  
},  
{  
  "id": 57,  
  "origin": 5,  
  "destination": 2,  
  "tick": 189
```

```
},  
{  
  "id": 58,  
  "origin": 1,  
  "destination": 4,  
  "tick": 190  
},  
{  
  "id": 59,  
  "origin": 5,  
  "destination": 2,  
  "tick": 194  
},  
{  
  "id": 60.
```

6. Tick和Event介绍

<https://zgca-forge.github.io/Elevator/events.html>

7. 类型参数和接口说明 `elevator_saga/client/proxy_models.py` 、 `elevator_saga/core/models.py`

<https://zgca-forge.github.io/Elevator/client.html>

8. HTTP API接口说明（给非Python实现的同学提供） `elevator_saga/server/simulator.py`

<https://zgca-forge.github.io/Elevator/communication.html>